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# TASK 1 — SQLAlchemy: Define Models and Connect
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from sqlalchemy import (
    create_engine, Column, Integer, String, Date, Numeric,
    ForeignKey, UniqueConstraint
)
from sqlalchemy.orm import declarative_base, relationship

engine = create_engine(

"postgresql+psycopg2://postgres:your_password@localhost:5432/college_db_orm",
    echo=True
)

Base = declarative_base()


class Department(Base):
    __tablename__ = "departments"

    dept_id = Column(Integer, primary_key=True, autoincrement=True)
    dept_name = Column(String(100), nullable=False, unique=True)
    head_of_dept = Column(String(100), nullable=False)
    location = Column(String(100))
    budget = Column(Numeric(12, 2))

    students = relationship("Student", back_populates="department")
    courses = relationship("Course", back_populates="department")
    professors = relationship("Professor", back_populates="department")


class Student(Base):
    __tablename__ = "students"

    student_id = Column(Integer, primary_key=True, autoincrement=True)
    first_name = Column(String(50), nullable=False)
    last_name = Column(String(50), nullable=False)
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    email = Column(String(100), nullable=False, unique=True)
    dob = Column(Date, nullable=False)
    dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
    enrollment_year = Column(Integer)

    department = relationship("Department", back_populates="students")
    enrollments = relationship("Enrollment", back_populates="student")


class Course(Base):
    __tablename__ = "courses"

    course_id = Column(Integer, primary_key=True, autoincrement=True)
    course_name = Column(String(150), nullable=False)
    credits = Column(Integer, nullable=False)
    dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
    max_seats = Column(Integer, default=60)

    department = relationship("Department", back_populates="courses")
    enrollments = relationship("Enrollment", back_populates="course")


class Professor(Base):
    __tablename__ = "professors"

    professor_id = Column(Integer, primary_key=True, autoincrement=True)
    first_name = Column(String(50), nullable=False)
    last_name = Column(String(50), nullable=False)
    email = Column(String(100), nullable=False, unique=True)
    hire_date = Column(Date)
    dept_id = Column(Integer, ForeignKey("departments.dept_id"), nullable=False)
    salary = Column(Numeric(10, 2))

    department = relationship("Department", back_populates="professors")


class Enrollment(Base):
    __tablename__ = "enrollments"
    __table_args__ = (UniqueConstraint("student_id", "course_id"),)
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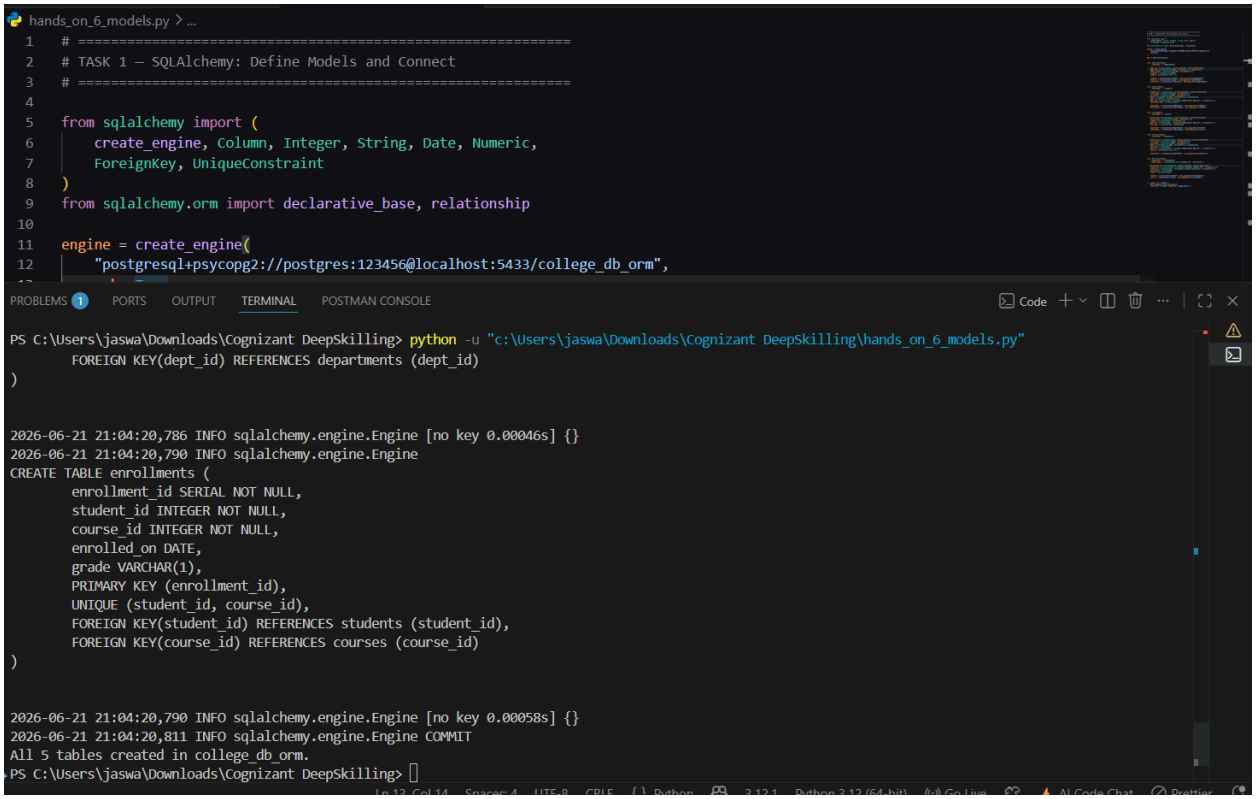
    enrollment_id = Column(Integer, primary_key=True, autoincrement=True)
    student_id = Column(Integer, ForeignKey("students.student_id"), nullable=False)
    course_id = Column(Integer, ForeignKey("courses.course_id"), nullable=False)
    enrolled_on = Column(Date)
    grade = Column(String(1))

    student = relationship("Student", back_populates="enrollments")
    course = relationship("Course", back_populates="enrollments")


if __name__ == "__main__":
    Base.metadata.create_all(engine)
    print("All 5 tables created in college_db_orm.")
```



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# ============================================================
# Query count comparison (documented as required by Step 90):
#
#   Task 2 Step 84 (lazy loading, default):
#       Querying all enrollments and accessing enrollment.student.first_name
#       and enrollment.course.course_name for each row triggers N+1 queries:
#       1 query to fetch all enrollments, then 1 additional query per
#       enrollment to lazy-load its related student, and 1 more per
#       enrollment to lazy-load its related course.
#       With 4 enrollments inserted in Step 82: 1 + 4 + 4 = 9 queries.
#       At larger scale (e.g. 13 enrollments from earlier hands-ons):
#       1 + 13 + 13 = 27 queries.
#
#   Task 3 Step 88 (joinedload, eager loading):
#       session.query(Enrollment).options(
#           joinedload(Enrollment.student),
#           joinedload(Enrollment.course)
#       ).all()
#       issues exactly 1 SQL query with JOINs, regardless of row count.
#
#   Result: query count drops from 9 (or 27) down to 1 — eliminating N+1.
# ============================================================

from datetime import date
from sqlalchemy import create_engine
from sqlalchemy.orm import sessionmaker, joinedload
from models import Base, Department, Student, Course, Enrollment, Professor

engine = create_engine(

"postgresql+psycopg2://postgres:your_password@localhost:5432/college_db_orm",
    echo=True
)

Session = sessionmaker(bind=engine)
session = Session()


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# TASK 2 — CRUD Operations via ORM
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departments = [
    Department(dept_name="Computer Science", head_of_dept="Dr. Alan Turing",
               location="Block A", budget=750000.00),
    Department(dept_name="Mathematics", head_of_dept="Dr. Emmy Noether",
               location="Block B", budget=620000.00),
    Department(dept_name="Electronics", head_of_dept="Dr. Nikola Tesla",
               location="Block C", budget=580000.00),
]
session.add_all(departments)
session.commit()

cs_dept = session.query(Department).filter_by(dept_name="Computer
Science").first()
math_dept = session.query(Department).filter_by(dept_name="Mathematics").first()
ec_dept = session.query(Department).filter_by(dept_name="Electronics").first()

students = [
    Student(first_name="Alice", last_name="Johnson",
email="alice.johnson@college.edu",
            dob=date(2001, 3, 15), dept_id=cs_dept.dept_id, enrollment_year=2021),
    Student(first_name="Bob", last_name="Smith", email="bob.smith@college.edu",
            dob=date(2000, 7, 22), dept_id=cs_dept.dept_id, enrollment_year=2021),
    Student(first_name="Carol", last_name="Williams",
email="carol.williams@college.edu",
            dob=date(2002, 1, 10), dept_id=math_dept.dept_id, enrollment_year=2022),
    Student(first_name="David", last_name="Brown",
email="david.brown@college.edu",
            dob=date(2001, 11, 5), dept_id=math_dept.dept_id, enrollment_year=2022),
    Student(first_name="Eva", last_name="Davis", email="eva.davis@college.edu",
            dob=date(2002, 6, 30), dept_id=ec_dept.dept_id, enrollment_year=2022),
]
session.add_all(students)
session.commit()

courses = [
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    Course(course_name="CS101 - Intro to Programming", credits=4,
dept_id=cs_dept.dept_id),
    Course(course_name="CS201 - Data Structures", credits=4,
dept_id=cs_dept.dept_id),
    Course(course_name="MA101 - Calculus I", credits=3,
dept_id=math_dept.dept_id),
]
session.add_all(courses)
session.commit()

alice = session.query(Student).filter_by(email="alice.johnson@college.edu").first()
bob = session.query(Student).filter_by(email="bob.smith@college.edu").first()
carol = session.query(Student).filter_by(email="carol.williams@college.edu").first()

cs101 = session.query(Course).filter_by(course_name="CS101 - Intro to
Programming").first()
cs201 = session.query(Course).filter_by(course_name="CS201 - Data
Structures").first()
ma101 = session.query(Course).filter_by(course_name="MA101 - Calculus I").first()

enrollments = [
    Enrollment(student_id=alice.student_id, course_id=cs101.course_id,
            enrolled_on=date(2021, 8, 1), grade="A"),
    Enrollment(student_id=alice.student_id, course_id=cs201.course_id,
            enrolled_on=date(2021, 8, 1), grade="B"),
    Enrollment(student_id=bob.student_id, course_id=cs101.course_id,
            enrolled_on=date(2021, 8, 1), grade="B"),
    Enrollment(student_id=carol.student_id, course_id=ma101.course_id,
            enrolled_on=date(2022, 8, 1), grade="A"),
]
session.add_all(enrollments)
session.commit()

cs_students = (
    session.query(Student)
    .join(Department)
    .filter(Department.dept_name == "Computer Science")
    .all()
)
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print("\n--- Students in Computer Science ---")
for s in cs_students:
    print(f"{s.first_name} {s.last_name}")

print("\n--- Enrollments (lazy loading — watch the SQL log for N+1) ---")
all_enrollments = session.query(Enrollment).all()
for e in all_enrollments:
    print(f"{e.student.first_name} {e.student.last_name} -> {e.course.course_name}")

student_to_update =
session.query(Student).filter_by(email="bob.smith@college.edu").first()
student_to_update.enrollment_year = 2023
session.commit()
print(f"\nUpdated {student_to_update.first_name}'s enrollment_year to
{student_to_update.enrollment_year}")

enrollment_to_delete = (
    session.query(Enrollment)
    .filter_by(student_id=carol.student_id, course_id=ma101.course_id)
    .first()
)
session.delete(enrollment_to_delete)
session.commit()
print("Deleted Carol's MA101 enrollment.")

remaining = session.query(Enrollment).count()
print(f"Remaining enrollments: {remaining}")


# ============================================================
# TASK 3 — Eager Loading to Fix N+1
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print("\n--- Enrollments (eager loading via joinedload — should be 1 query) ---")
eager_enrollments = (
    session.query(Enrollment)
    .options(
        joinedload(Enrollment.student),
        joinedload(Enrollment.course)
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    )
    .all()
)
for e in eager_enrollments:
    print(f"{e.student.first_name} {e.student.last_name} -> {e.course.course_name}")

session.close()
```